Roll No……..

ODD END SEMESTER EXAMINATION DECEMBER – 2025

Name of the Course: M.Tech (ME)

Semester: III

Name of the Paper: Computer Aided Analysis of Mechanical System

Paper Code: DEME 53

Time: 3 Hours

Maximum Marks: 100

Note:

(i) All questions are compulsory.
(ii) Answer ***any two sub questions*** among a, b, & c in each main question.
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks.

Q1	(20 marks)	**CO 1**
(a)	Discuss how computers assist in solving eigenvalue problems in mechanical design?	
(b)	Explain the significance of system modeling and representation in dynamic design?	
(c)	A 2-DOF mass–spring system has stiffness matrix [k] = [[4000, -2000], [-2000, 4000]] N/m and mass matrix [m] = [[2, 0], [0, 1]] kg. Determine the natural frequencies?	
Q2	(20 marks)	**CO 1 & 2**
(a)	Describe the steps involved in the computer-aided design process for dynamic systems?	
(b)	Derive the equation of motion for an SDOF system subjected to harmonic excitation?	
(c)	Explain the difference between free vibration and forced vibration responses in mechanical systems?	
Q3	(20 marks)	**CO 2**
(a)	Discuss the concept of hysteric damping with a suitable example?	
(b)	A 1-DOF system has m = 4 kg, k = 1600 N/m, and damping ratio ζ = 0.1. Determine the damped natural frequency and logarithmic decrement?	

(c)	Explain the experimental techniques used for vibration testing of mechanical systems?	
Q4	(20 marks)	**CO 3 & 4**
(a)	Describe various types of excitation used in structural testing and their advantages?	
(b)	Discuss the function of amplifiers and analyzers in dynamic signal measurement?	
(c)	Explain the use of digital signal processing (DSP) in vibration analysis?	
Q5	(20 marks)	**CO 4 & 5**
(a)	Write a detailed note on – (a) Multi-point excitation methods, and (b) Mass cancellation technique?	
(b)	For a system, the transfer function is $H(\omega) = 1 / (k - m\omega^2 + j\, c\, \omega)$. Derive the magnitude and phase response and sketch their variation with frequency?	
(c)	Explain the differences among modal models, response models, and spatial models in mechanical system analysis?	